

Padding Oracle Attack

Understand and demonstrate how a Padding Oracle Attack can recover plaintext from an AES-CBC encrypted message without knowing the encryption key.

Scenario:

Lets assume you have:

- An AES-CBC encrypted ciphertext (You may use a standard AES cryptographic library for encryption/decryption operations)
- IV
- A padding-oracle function that returns whether the decrypted ciphertext has valid PKCS#7 padding
- Starter code for AES-CBC encryption and the oracle

The AES key will not be provided.

Tasks

1. **Understand the Attack**
Briefly explain AES-CBC, PKCS#7 padding, and the role of a padding oracle.
2. **Implement the Attack**
Complete the given starter code to:
 - Modify the previous ciphertext block.
 - Query the padding oracle.
 - Recover plaintext bytes from **right to left**.
 - Recover the complete plaintext.
3. **Analyze the Attack**
Record the total number of oracle queries required and explain why modifying the previous ciphertext block affects the plaintext of the target block.
4. **Security Recommendation**
Explain how a padding oracle vulnerability can be prevented in a real system.

Constraints

- The AES key must **not be used or accessed** during the attack.
- Do not use a ready-made padding-oracle attack library.
- The solution must work for the given ciphertext and not be hard-coded to a particular plaintext.

Deliverables

- Completed Python program
- Recovered plaintext
- Number of oracle queries
- Brief explanation of the attack and its prevention